

Agent360: A Concealment-First Bilateral Negotiator for ANL 2026

Noam Kazum
The College of Management Academic Studies

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Abstract

We present **Agent360**, a bilateral negotiator for the ANL 2026 league that maximizes $\text{Score} = \text{Advantage} + \text{Concealing}$. The agent combines a *decoy persona*—early bids that misrepresent which issues we value—with a deception-aware opponent model built only from observed offers, seat order, and relative time. A phased bidding strategy (decoy, transition, closing) protects Concealing, while mode-dependent acceptance and Smith-based closing bids improve Advantage against conceders, learners, and deceptive opponents. Development used iterative local benchmarks and official tournament feedback; we found that small acceptance tweaks aimed at reducing no-deal outliers can hurt overall performance if validated only on proxy panels rather than the full opponent mix.

1 Introduction

The ANL 2026 competition rewards negotiators on two axes: **Advantage** (agreement utility minus reserved value) and **Concealing** (how poorly the opponent models our true preferences from our bid stream and published models). Agents that concede quickly or reveal true issue priorities early often score well on Advantage against simple opponents but lose Concealing against Smith-style learners (e.g. BOA, MAP, MiCRO).

Agent360 is designed *concealment-first*: we mislead the opponent’s learner for as long as it remains profitable, then gradually shift toward profitable agreements. We use no oracle access to the opponent’s true utility or agent class—only observed offers, negotiation seat, and relative deadline time.

2 The Design of MyAgent

Agent360 extends the NegMAS SAOCallNegotiator protocol with three cooperating layers:

1. **Bidding persona:** what we *offer*—decoy pool construction, phased transition, and closing bids.
2. **Opponent model:** what we *infer*—frequency (Smith), recency, time-weighted, and trajectory-based estimates plus behavioral mode classification.
3. **Deal extraction:** when we *accept*—time- and mode-dependent aspiration, bait detection, and deadline safeguards.

Each step with a partner offer: update all models; evaluate acceptance; otherwise sample the next bid from the current phase. The first negotiator (seat 0) follows a stricter concealment profile—longer decoy phase, a minimum count of opponent bids before transition, and decoy rotation with utility jumps—because opening agents expose more of their bid stream to opponents’ learners.

3 Concealing Bidding Strategy

Relative time $t \in [0, 1]$ divides the negotiation into three phases.

Decoy. We bid from a **maximal-mismatch decoy pool**: rational outcomes that disagree with our inferred true top-issue values on at least half of issues, while staying above a utility floor. When we open, transition is blocked until the opponent has made at least four offers. Decoy selection avoids recent repeats, breaks mirror tit-for-tat, and prefers bids with large utility jumps relative to our last offer.

Transition ($t < 0.75$). We mix decoy outcomes with a widening band of our true high-utility outcomes. The first negotiator keeps decoy mixing for most of the transition window and reveals true preferences more slowly than the second seat.

Closing ($t \geq 0.75$). We sample rational outcomes and score each by a blend of normalized self-utility and estimated opponent utility. The weight on the opponent model increases through closing and is capped by the classified opponent mode (higher vs. learners and conceders, lower vs. deceptive or mirroring opponents).

Together, decoy persona, gradual transition, and opponent-weighted closing form our main Concealing tactic: frequency learners fit an incorrect early preference signal.

4 Acceptance Strategy

We accept when any of the following applies:

- **Catastrophe floor:** $t \geq 0.95$ and utility $\geq$ reserved value.
- **Mode aspiration:** utility $\geq u_{\max}(1 - \text{slope} \cdot t)$ and above reserved value; aspiration slope is tuned per opponent mode.
- **AC-next:** offer utility is at least that of our next concealing bid.
- **Deadline safe:** $t > 0.90$ and utility above reserved value.

Before the deadline-safe window, offers that pass aspiration may still be rejected if they appear to be **bait**: a deceptive opponent profile and Smith-estimated utility far above the trajectory prediction. During development we experimented with earlier escape acceptance (e.g. accepting above reserved value well before the deadline); this improved some local stress tests but was dropped because it accepted too many weak or deceptive deals in broader evaluation.

5 Opponent Model

All inference uses opponent bids only:

- **Smith frequency model** over the full history.
- **Recency-weighted Smith** on the second seat (recent window of five offers).
- **Time-weighted Smith** emphasizing bids after $t = 0.40$.
- **Trajectory model** of $(t, \hat{u}_{\text{Smith}})$ for concession slope, non-monotonicity, and bait inconsistency.

We classify opponents as `mirror`, `learner`, `deceptive`, `conceding`, or `unknown` from bid concentration, early issue-flip rate, trajectory shape, and bid-echo detection—not from agent class names. The published `opponent_ufun` blends these estimators; against deceptive late bait-switching we discount inflated Smith scores when choosing closing bids and when deciding acceptance.

6 Evaluation

We evaluated Agent360 locally before each submission using a repeatable panel:

- **Learners:** BOA, MAP, MiCRO across all competition scenarios (both seats).
- **Stress:** time-based NegMAS negotiators (Boulware, Conceder, Tough, etc.).
- **Sparring:** in-house deceptive opponents (mirror, bait-switch, strong learner variants).

Each cell is repeated with fixed step counts; we report mean Advantage, Concealing, and their sum.

Table ?? shows a representative snapshot on the combined panel. Official tournament runs use a different opponent mix; we therefore treated leaderboard feedback as the final gate for acceptance-rule and closing-weight changes, not proxy means alone.

Table 1: Local proxy panel (mean Score per matchup cell; illustrative development snapshot).

Opponent group	Advantage	Concealing	Score
Learners (BOA/MAP/MiCRO)	0.56	0.71	1.27
Stress (time-based)	0.53	0.90	1.43
Deceptive sparring	0.61	0.58	1.19
Panel mean	0.55	0.79	1.34

Iterative changes that improved the submitted agent include: conceding-triggered early exit from decoy, relaxed bait discount thresholds, mode-dependent closing caps, first-seat decoy rotation, anti-mirror bid pools, and catastrophe acceptance near timeout. Changes we tried but did not keep include stricter first-seat min-offer gates, shorter second-seat decoy windows, and mid-game escape acceptance—they helped isolated tail cases on local runs but reduced robustness on the wider panel.

7 Lessons and Suggestions

Separate concealment from deal quality. Concealing is driven mainly by the bid stream; Advantage depends on opponent modeling and acceptance. Strengthening one without weakening the other requires keeping the decoy persona while tuning closing and acceptance independently.

Local benchmarks are necessary but not sufficient. Proxy panels catch regressions against known opponent families but do not replicate the full tournament distribution. Major design decisions should be checked on official runs when available.

Seat asymmetry matters. First and second negotiators see different information; asymmetric decoy length and min-offer gates improved learner matchups in our local tests without oracle routing.

Avoid over-fitting to tail outcomes. Rules that accept earlier to prevent timeouts can trade away Advantage and Concealing on typical head-to-head negotiations.

Conclusions

Agent360 is a concealment-first bilateral negotiator that combines maximal-mismatch decoy bidding, gradual preference revelation, deception-aware Smith modeling, and mode-tuned acceptance and closing. It operates without oracle information and adapts behavior to negotiation seat and inferred opponent type. Our development process emphasized iterative local evaluation followed by tournament-informed refinement, with particular care not to sacrifice the decoy persona when improving deal extraction.